

SINVERT PVM

Operating Instructions · 07/2010



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SINVERT PVM

Operating Instructions

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Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

DANGER

indicates that death or severe personal injury **will** result if proper precautions are not taken.

WARNING

indicates that death or severe personal injury **may** result if proper precautions are not taken.

CAUTION

with a safety alert symbol, indicates that minor personal injury can result if proper precautions are not taken.

CAUTION

without a safety alert symbol, indicates that property damage can result if proper precautions are not taken.

NOTICE

indicates that an unintended result or situation can occur if the corresponding information is not taken into account.

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The product/system described in this documentation may be operated only by **personnel qualified** for the specific task in accordance with the relevant documentation for the specific task, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

Proper use of Siemens products

Note the following:

WARNING

Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be adhered to. The information in the relevant documentation must be observed.

Trademarks

All names identified by ® are registered trademarks of the Siemens AG. The remaining trademarks in this publication may be trademarks whose use by third parties for their own purposes could violate the rights of the owner.

Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

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Introduction

1.1 About this manual

Trademarks

SINVERT® is a registered trademark of Siemens AG.

Purpose of this manual

These operating instructions contain all the information required to install, commission, and operate SINVERT PVM10, SINVERT PVM13, SINVERT PVM17 4DC, SINVERT PVM17 6DC, and SINVERT PVM20 inverters.

The SINVERT PVM17 4DC inverter is no longer available to order.

Basic knowledge required

- Training as an electrical technician
- Experience of working with PV systems
- Experience of installing and commissioning PV systems
- Experience of working with inverters

Conventions

In this manual, the abbreviation SINVERT PVM is also used to refer to inverters of type SINVERT PVM10, SINVERT PVM13, SINVERT PVM17 4DC, SINVERT PVM17 6DC, and SINVERT PVM20.

The latest version of the operating instructions is available on the Internet.

These operating instructions are regularly updated. For this reason, always download the latest version from the Internet so that you always have the valid operating instructions for any variant of the SINVERT PVM inverter.

Follow the link below to access the instructions:

SINVERT support (<http://www.siemens.com/sinvert-support>)

1.2 History of changes in this documentation

Release	Remark
12/2009	First edition (German only)
01/2010	Changes (German only)
03/2010	Changes (German and English)
07/2010	Addition of the SINVERT PVM17 6DC and SINVERT PVM20 inverters; changes

Safety instructions

2

To avoid personal injury or material damage, the following instructions must be read before commissioning the unit for the first time. These safety instructions must be followed at all times.

WARNING

Improper handling of these devices and failure to follow the warnings given here, as well as improper interference with the safety equipment and the unit, can lead to property damage, bodily injury, or electric shock, and in extreme cases can cause death.

WARNING

Risk of electric shock

Do not open the device! The unit interior can contain fatal voltage, even after it is switched off.

CAUTION

Safety hazards associated with using unsuitable PV modules

The SINVERT PVM inverter is not suitable for use with PV modules for which pole grounding is an essential requirement. Modules of this type must not be connected to the SINVERT PVM inverter. Failure to provide suitable grounding poses a safety risk to personnel and the installation itself.

CAUTION

Invalidation of the warranty

The device is only permitted to be opened by authorized specialist personnel, otherwise the warranty will be invalidated.

2.1 Qualified personnel

These operating instructions are intended for the following persons:

- Electrical technicians who commission the unit and connect it to other units in the PV system
- Service and maintenance engineers who are installing upgrades or performing error analyses

This documentation is written for trained specialists. It does not provide basic information about PV systems.

- Only trained specialists may install the unit.
 - The installation engineer must be qualified according to the national guidelines.
 - Approval by the relevant electrical utility may also be necessary.
- **Operation, maintenance, and repair of this device may be made only by qualified staff who are trained to work on or with electrical devices.**

See also

Technical support (Page 91)

2.2 Protection against touching electrical parts

WARNING

Risk of electric shock

- Before switching on, ensure the connectors are seated properly.
- Remove the PV generator connectors only when the following conditions have been met:
 - Switch the DC disconnect on the SINVERT PVM to "OFF".
 - Measure at the DC end to make sure that no current is flowing.
 - Disconnect the power supply and take measures to prevent the power being switched back on again.

WARNING

Electric shock and fire hazard due to high leakage current

Create a ground connection before connecting to the PV field and the power supply circuit.

2.3 Protection against magnetic and electromagnetic fields in operation and installation

The magnetic and electromagnetic fields that are present in the immediate vicinity of electrical conductors can represent a serious danger to people with pacemakers, metal implants, and hearing aids.

WARNING

People with pacemakers, metal implants, and hearing aids are at risk of damaging their health in the immediate vicinity of electrical equipment.

- Access to the following areas is prohibited for people with pacemakers and metal implants:
 - Areas in which electrical devices and components are installed, operated, or commissioned.
- If it is necessary for a pacemaker wearer to enter such areas, this requires a doctor's prior approval. The resistance to interference of pacemakers that have been implanted or are yet to be implanted varies greatly, and general guidelines are therefore not possible.
- People with metal implants or fragments and those with hearing aids must consult a doctor before entering such areas, as dangers to health can be expected.

2.4 Protection against burn hazards

CAUTION

Surfaces of device enclosure may be hot! Risk of injury! Hot parts – burn hazard!

- Do not touch cooling fins and the upper area of the enclosure during operation. Hot parts – burn hazard!
- Before touching the unit, let it cool down for 15 minutes.
- At an ambient temperature of 45 °C, the upper part of the enclosure and the heat sink can assume a surface temperature of 75 °C!

2.5 Protection during handling and installation

Improper handling and installation of certain parts and components can result in injury under unfavorable conditions.



CAUTION

Danger of injury due to improper handling! Injury by crushing, jackknifing, cutting, bumping, or lifting!

- The general construction and safety regulations must be observed in handling and installation.
- A SINVERT PVM inverter weighs approximately 40 kg.
- Suitable installation and transport equipment must be used.
- Only use suitable tools.
- Lifting equipment and tools must be used correctly.
- Suitable protective equipment (e.g. safety goggles, safety shoes, protective gloves) must be used.
- Never stand underneath suspended loads.

2.6 Selecting the country setting

The country setting determines the country-specific line monitoring parameters. In selecting the country setting, the menu language is set automatically. The menu language is then independent of the country setting and can be freely selected at any time using the menu.

No country setting is selected when the device is delivered. The device prompts you to select the country setting. You can choose among the countries listed. The term "country setting" itself does not appear in the menu.

CAUTION

The selected country setting can be changed by service personnel only!

After selecting and confirming the country setting, you can no longer change the country setting yourself. According to the regulations, the country setting can be changed by service personnel only.

If the unit has been in operation, the country setting is already selected. In this event, you will not be asked to enter the country setting. The country setting can be changed by service personnel only.

CAUTION

Withdrawal of operating permit

If you operate SINVERT PVM with an incorrect country setting, the electrical utility can withdraw your operating permit.

We assume no responsibility for an incorrect country setting.

Description

3.1 Device description

SINVERT PVM is a transformerless, three-phase inverter with the following features:

- Output and efficiency
 - SINVERT PVM works with particularly high efficiency in every operating point.
 - SINVERT PVM is suitable for connecting a PV generator with a power of up to 21 kWp.
- Design features
 - The device does not have to be opened for mounting and connecting.
 - All electrical connections have lockable connectors.
 - SINVERT PVM has an integral DC disconnecter in accordance with EN 60947-3 that substantially reduces the overall overhead of the installation engineer.
- Heat dissipation
 - Heat is dissipated purely by means of convection.
 - An internal temperature monitor protects the device against exceeding the permissible ambient temperature.
- Interfaces
 - Ethernet
 - RS485
 - USB
 - Relay 230 V / AC 4 A for fault signaling
- Operator input
 - An illuminated graphics display clearly represents the waveform of the infeed power and other data.
 - Operator input is via keys below the display.
- Area of application

Thanks to its IP65 degree of protection, the SINVERT PVM inverter can be used outdoors almost without restrictions.



Figure 3-1 SINVERT PVM

3.2 Scope of supply

The scope of supply of the SINVERT PVM inverters includes the following:

- SINVERT PVM inverter:
SINVERT PVM10 or SINVERT PVM13 or SINVERT PVM17 or SINVERT PVM20
- Wall bracket
- Accessory pack
 - 3 x flat washer form B M8 DIN125-8 for wall bracket
 - 2 x flat-head screw M5x20,
for mechanically securing the inverter in the wall bracket
 - 1 x contact insert IP67 5-pin VC-TFS5-PEA
 - 1 x socket shell IP67 VC-K-T3-R-M25-PLOMB
 - 1 x cable gland Schlemmer-Tec M25x1.5/21532
 - 2 x connector 4-pin SACC-M12MS-4SC for RS485 interface
 - 1 x socket 4-pin SACC-M12MS-4SC for relay connection
 - 1 x Ethernet connector Phoenix Contact VS-08-RJ45-5-Q / IP67
- Operating instructions (compact) as hard copy
- CD with operating instructions

Note

Spare parts

The wall bracket and the accessory pack with the connector set can be ordered separately as spare parts. See section Spare parts and accessories (Page 89)

3.3 External device dimensions

All dimensions are in millimeters.

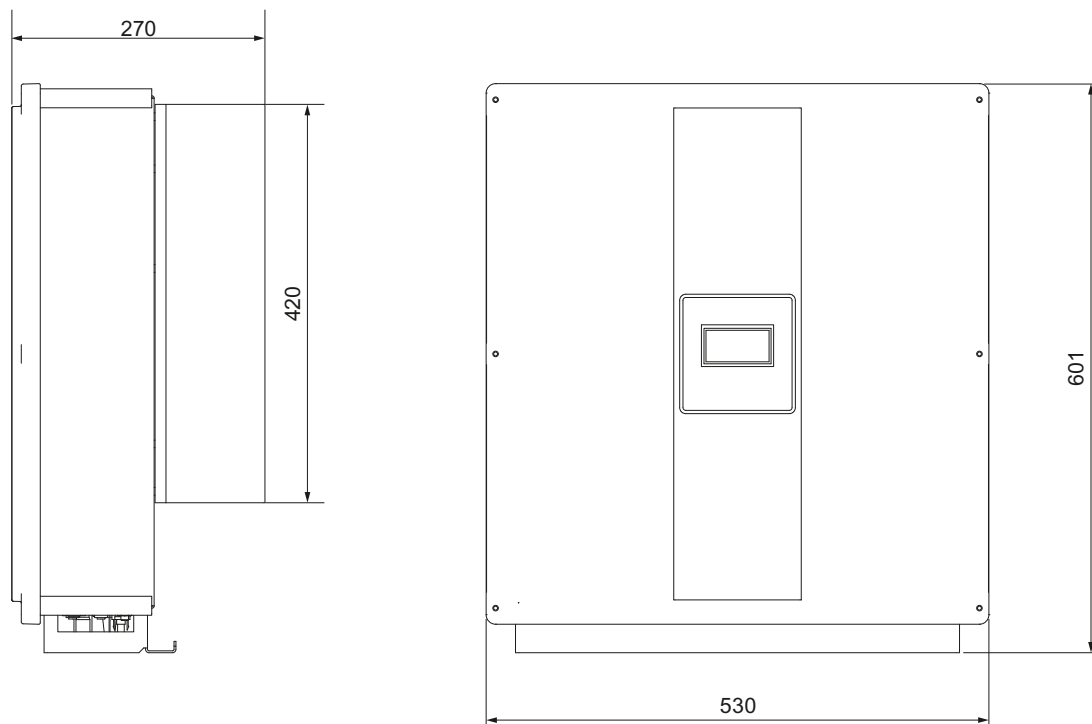


Figure 3-2 Dimension drawing

3.4 Block diagram

The SINVERT PVM inverters are equipped with a surge arrester of Category 3 both on the AC side and the DC side.

Block diagram for SINVERT PVM10, SINVERT PVM13, and SINVERT PVM17 4DC ¹⁾

¹⁾ SINVERT PVM17 4DC inverter is no longer available to order.

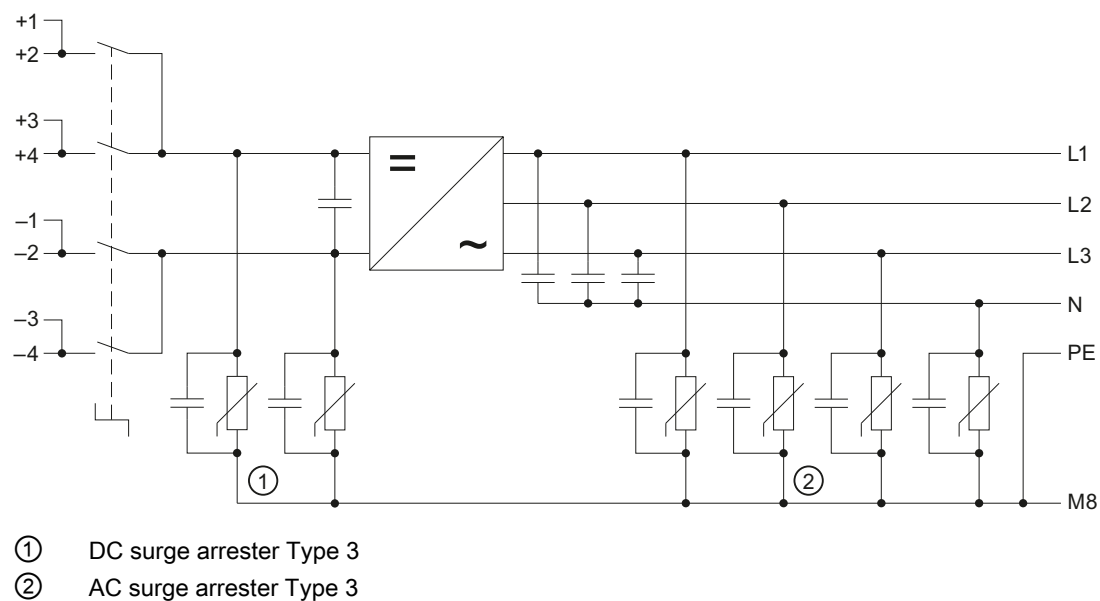


Figure 3-3 Block diagram with four DC inputs

Block diagram for SINVERT PVM17 6DC and SINVERT PVM20

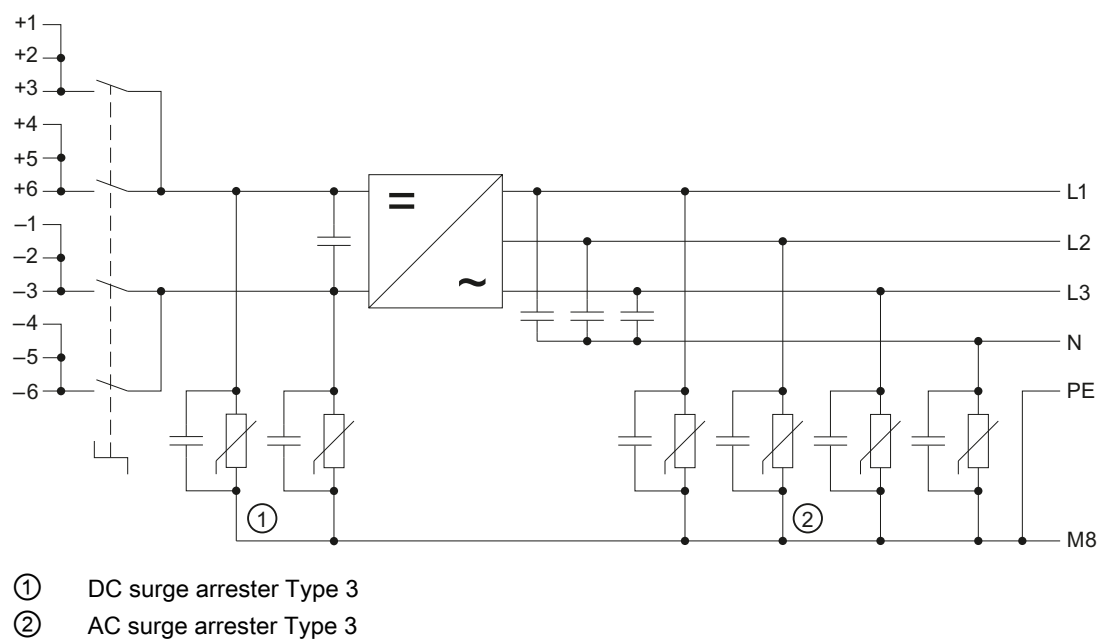


Figure 3-4 Block diagram with six DC inputs

3.5 Operator panel

The waveform of data such as the infeed power can be represented with the 128 x 64-pixel graphics display integrated on the front. The necessary parameters are selected and entered using the 8-key operator panel. The display is illuminated when a key is first pressed. The lighting goes out automatically after a few seconds.

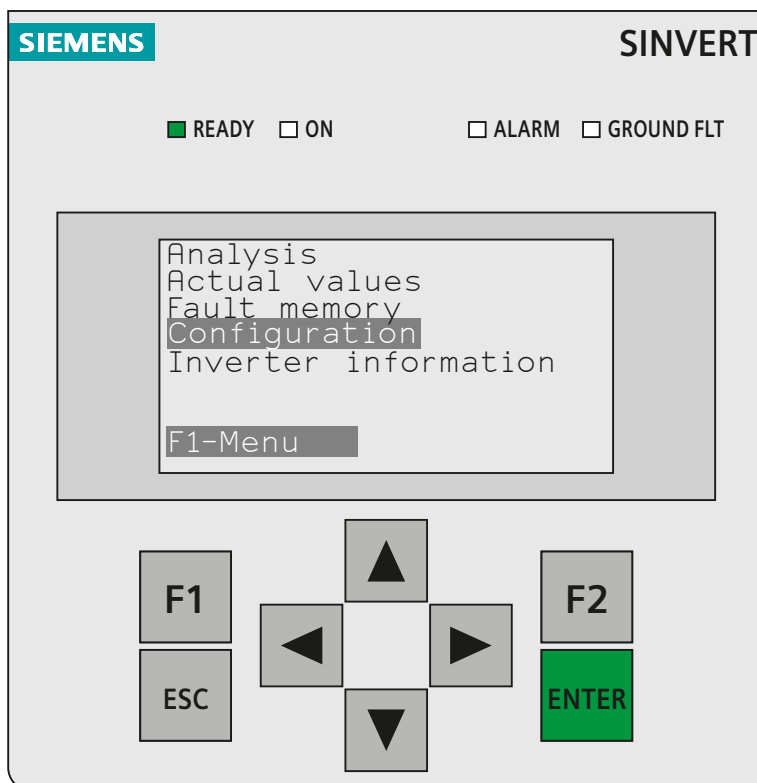


Figure 3-5 Operator panel

Status display

Designation	Description
READY	LED flashing: Initialization LED is lit: Initialization of the unit is complete, but the inverter itself is not yet switched in. The inverter is not yet drawing current. See Section Activating the unit (Page 56) for more details
ON	LED flashing: Switching in the inverter LED is lit: Infeed operation See Section Activating the unit (Page 56) for more details
ALARM	LED flashes red: Transient fault. The error message is saved in the fault memory, safe from power outages. See section Transient fault (Page 81) for more details LED shows red light: Fault: The error message is saved in the fault memory, safe from power outages. See section Faults (Page 82) for more details
GROUND FLT	The insulation value in the PV field is too low. See section Isolation faults (Page 82)

Display

- Depending on the selection, the display shows the following information:
 - Status display
 - Infeed power
 - Energy data
 - Device menu
- Selection via the input keys.
- The display is illuminated as soon as a key is pressed and it switches off automatically.

Input keys

Designation	Description
F1	Calling up the device menu
F2	Select entry of normalized energy data
↑ ↓	Selection
← →	Function in the menu: Jump to first or last menu item Function on parameter change: Position left, position right (skip screen)
ESC	Fault acknowledgement, entry deletion
ENTER	Confirmation of a selection or entry

3.6 Internal datalogger

The SINVERT PVM inverter contains an internal datalogger that enables the parallel recording of 40 measured values in the form of parameters. The values can be stored for up to approximately 14 years, depending of the recording cycle selected.

Recording cycle	Storage time
Every minute	Approx. 8.5 months
Every 10 minutes	Approx. 7 years
Every 20 minutes	Approx. 14 years

The data logger is designed as a ring buffer. If the memory capacity is exceeded, the oldest measured values are overwritten.

Description

3.6 Internal datalogger

Application planning

4.1 Requirements to be met by the PV field

The following operating data must on no account be exceeded by the PV generator!

Device type	SINVERT				
	PVM10	PVM13	PVM17 4DC ¹⁾	PVM17 6DC	PVM20
Maximum DC voltage per input	1000V				
Maximum DC current per switch contact (see Section Block diagram (Page 19))	25 A				
Maximum DC current at the input across all connections ²⁾	29 A	30 A	32 A	37 A	41 A

¹⁾ No longer available to order

²⁾ See below for the connection method.

 WARNING
The plus or minus pole of the PV generator must never be grounded.

Method of connection

- The power of the PV field must be distributed across all inputs of the inverter as evenly as possible.

WARNING

The plus or minus pole of the PV field must never be grounded.

SINVERT PVM10 and SINVERT PVM13 and SINVERT PVM17 4DC inverters

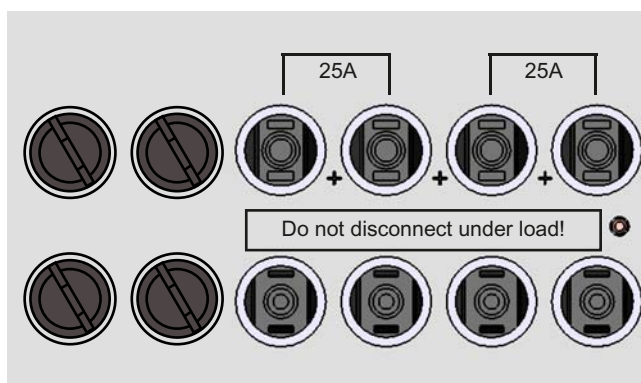


Figure 4-1 PV field connection with four DC inputs

- SINVERT PVM10, SINVERT PVM13, and SINVERT PVM17 4DC (no longer available to order) inverters each have four DC inputs for the PV field.
- If necessary, the connection can also be distributed across two inputs as follows.
In this case you can use inputs 1 and 3 or inputs 2 and 4.

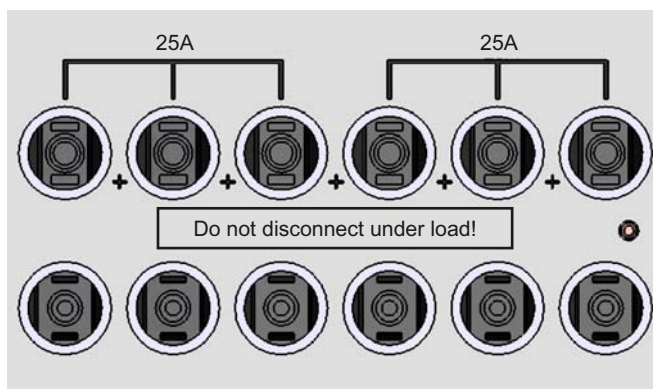
SINVERT PVM17 6DC and SINVERT PVM20 inverters

Figure 4-2 PV field connection with six DC inputs

- SINVERT PVM17 6DC and SINVERT PVM20 inverters each have six DC inputs for the PV field.
- If necessary, the connection can also be distributed across two inputs as follows.
In this case you can use inputs 1 and 4 or inputs 2 and 5 or inputs 3 and 6.

4.2 Reverse current

Reverse currents are fault currents that can occur in PV systems consisting of lines switched in parallel.

Reverse currents can have the following causes:

- Short-circuits of individual modules
- Short-circuits of cells in a module
- Double ground fault

As a result, the open-circuit voltage of the affected string can decrease (e.g. through defective modules or parts of modules) to such an extent that the intact parallel-switched strings cause a reverse current through the defective string. This can destroy the string in the worst case.

The sometimes intense overheating of the damaged string caused by the reverse current can also result in secondary damage.

To avoid such damage to PV systems, appropriate precautions must be taken. We must distinguish between two cases here:

1. The PV system is designed in such a way that the reverse current flowing in the event of a fault results in neither destruction of the damaged lines nor in any secondary damage. In the worst case, the reverse current comprises the sum of the short-circuit currents of all intact lines. The current-carrying capacity of the plant components (connectors, cables) and the reverse current resistance of the modules are decisive here. Please refer to the manufacturer's data sheet for the current-carrying capacity.
2. The PV system is designed in such a way that the reverse current occurring in the event of a fault exceeds the destruction limit. In this case, each string must be fused individually using a string fuse switched in series. In the event of a fault, the line is isolated from the intact lines and destruction is prevented.

NOTICE
Please note that fuses may be required in the module data sheet.

4.3 Communication

The Ethernet interface and the RS485 interfaces are used for remote monitoring of the SINVERT PVM inverters:

- Remote monitoring via SINVERT Webmonitor
- Standardized setting of setpoints for certain system configurations via the SINVERT PVM Control Box

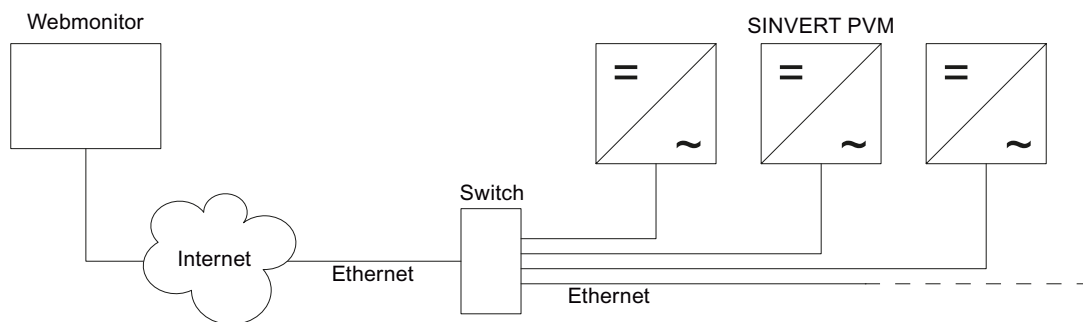


Figure 4-3 Communication via Ethernet

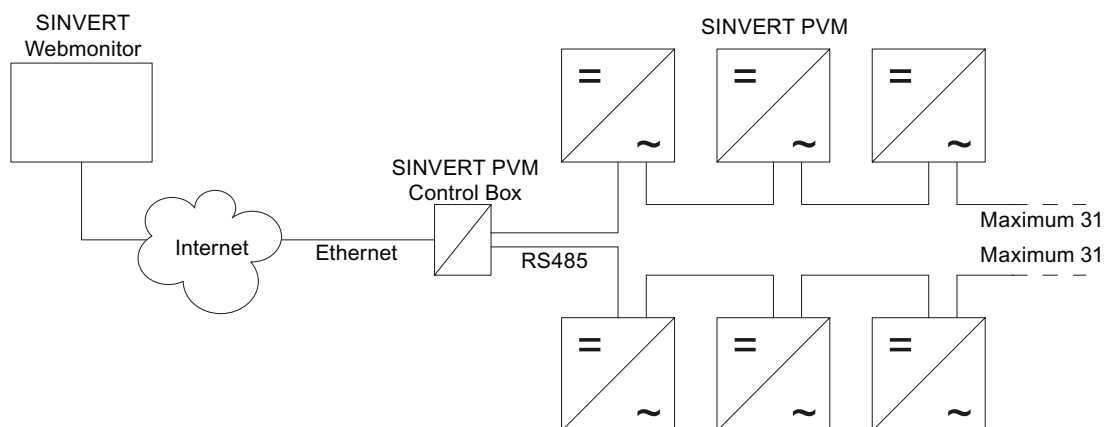


Figure 4-4 Communication via RS485 interface

Up to 31 devices can be connected to one RS485 bus.

4.4 Infeed into the AC system

NOTICE

Possible EMC problems caused by inadequate grounding

In order to satisfy EMC standards, ground the inverter at the M8 grounding point with at least 10 mm²!

Infeed into the low-voltage system

TN-S system

In a TN-S system, the protective conductor PE is connected separately. Loads can be connected directly to PE.

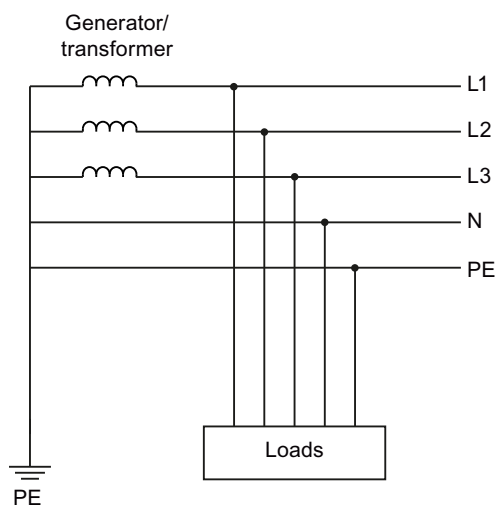


Figure 4-5 TN-S system

TT system

In a TT system, there is no separate protective conductor PE. The load must be connected via ground.

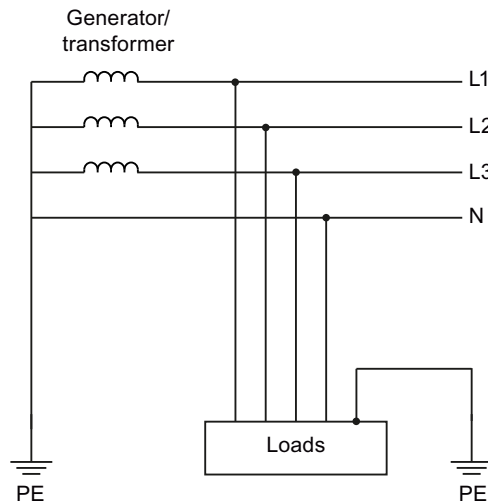


Figure 4-6 TT system

When it is connected to a TT system without a separate protective conductor PE, a separate PE cable with a cross-section of at least 10 mm² must be connected to and grounded on the inverter.

Infeed management

The latest amendment to the Renewable Energy Sources Act requires that from January 2009, PV systems with a connected load of more than 100 kWp must be included in infeed and system safety management.

The new BDEW guideline "Generating Plants in the Medium-Voltage Grid", which is in force in Germany, stipulates this requirement for all systems feeding in at medium-voltage levels. The primary significance of this ruling is that grid operators now have the option of limiting the power of the plant by remote control in accordance with §6 of the Renewable Energy Sources Act 2009.

Therefore, to use a PV system with a connected load of more than 100 kWp in Germany you need the SINVERT Control Box as an accessory.

Features of the SINVERT PVM Control Box

The features of the SINVERT PVM Control Box are as follows:

- The SINVERT PVM Control Box permits communication with up to 2 x 31 SINVERT PVM inverters.
- All of the connected SINVERT PVM inverters can be connected over an RS485 bus via the SINVERT PVM Control Box to the SINVERT Webmonitor
- The output power is limited in stages of 0%, 30%, 60%, and 100% by means of a ripple-control receiver connected to the SINVERT PVM Control Box. The power setpoints are transferred to the inverters over the RS485 bus.
- Cos ϕ specification as a fixed value: The SINVERT PVM Control Box distributes the specified cos ϕ setpoint to the individual SINVERT PVM inverters.
- The specified power limitation is stored in the SINVERT PVM Control Box.

4.5 Transportation

The devices must be transported in a clean and dry state, preferably in their original packaging. The transport temperature must be between - 25 °C and + 70 °C. Temperature fluctuations greater than 20 K per hour are not permitted.

4.6 Storage

The devices must be stored in clean and dry rooms, preferably in their original packaging. The storage temperature must be between - 25 °C and + 55 °C. Temperature fluctuations greater than 20 °C per hour are not permitted.

NOTICE
The SINVERT PVM inverter contains electrolytic capacitors. These capacitors can be stored at zero voltage for up to 2 years at a storage temperature of ≤ 40 °C.
If the storage period of two years is exceeded, please contact Service & Support before connecting the SINVERT PVM inverter to the system, see Technical support (Page 91).

5.1 Requirements regarding the installation location

The SINVERT PVM inverter uses pure convection cooling, and is therefore designed for mounting on a vertical wall. Installation is carried out by means of a wall bracket.

NOTICE
To avoid accidents during installation and service operations, there must be free and safe access to the units.

- The unit must be freely accessible.
- There must be no direct sunlight.
- Only vertical mounting is approved.
- Use a solid wall or metal structure. It must be ensured that module racks made of wood can bear the weight.
Please ensure the wall is strong enough. At least 40 kg per unit is required.
- Sufficient distance from flammable materials.
- For simple operation, an installation at chest level is recommended.
- Outdoor installation is also possible thanks to IP65 degree of protection.
- The rating plate must be readable.
- The connectors must remain accessible, particularly those for USB and Ethernet.

CAUTION
To ensure the IP65 degree of protection, the supplied connectors and sockets must be used to connect the SINVERT PVM inverter, following the connector manufacturer's enclosed installation instructions.
To protect against the ingress of moisture and dirt, unused inputs and outputs must be closed. Failure to follow this guideline may void the warranty!

CAUTION

Danger of overheating the units

- To avoid impairing convection cooling performance, SINVERT PVM inverters may not be mounted directly on top of one other. Make sure that the minimum distances are not exceeded.
- The cooling fins on the heat sink must never be covered. Failure to follow this guideline may void the warranty.

- To allow the necessary heat dissipation, the following minimum distances to the ceiling and wall, or adjacent units, must be observed.

Minimum distances:	
lateral	50 mm
top	500 mm
bottom	500 mm

5.2 Protection during handling and installation

Improper handling and installation of certain parts and components can result in injury under unfavorable conditions.



CAUTION

Danger of injury due to improper handling! Injury by crushing, jackknifing, cutting, bumping, or lifting!

- The general construction and safety regulations must be observed in handling and installation.
- A SINVERT PVM inverter weighs approximately 40 kg.
- Suitable installation and transport equipment must be used.
- Only use suitable tools.
- Lifting equipment and tools must be used correctly.
- Suitable protective equipment (e.g. safety goggles, safety shoes, protective gloves) must be used.
- Never stand underneath suspended loads.

5.3 Installation of SINVERT PVM inverter

A supplied wall bracket is used in installation.

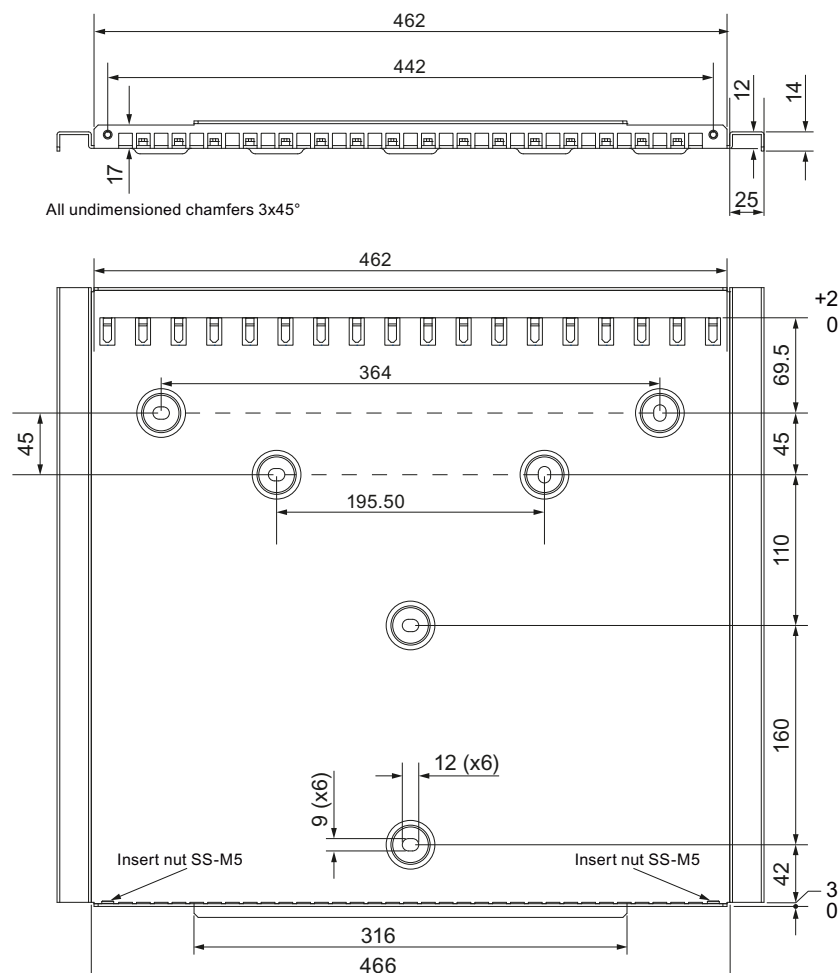


Figure 5-1 Wall bracket

CAUTION

Risk of injury

In setting up a wall mount, the weight of the SINVERT PVM inverter, around 40 kg, must be accounted for.

Failure to observe these requirements can result in a device malfunction or even in serious injury as a result of crushing, jackknifing, cutting, bumping and fire!

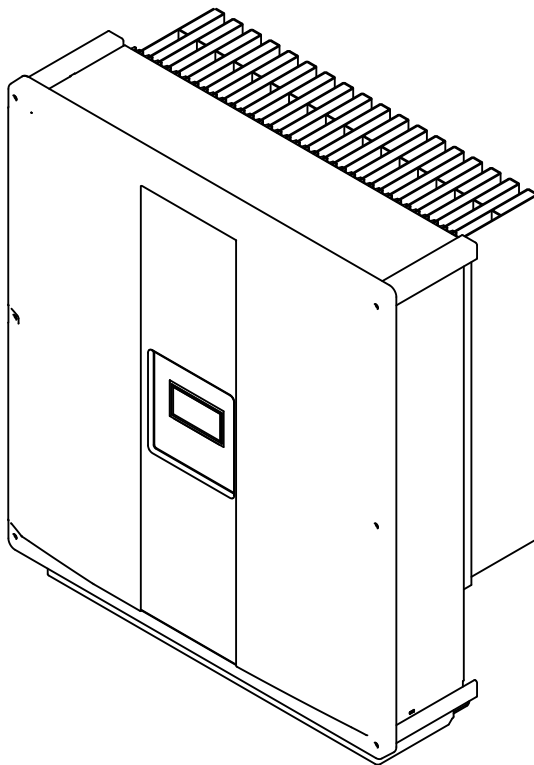


Figure 5-2 SINVERT PVM

1. Installation of wall bracket.
 - You can use the wall bracket as a stencil to mark the position for the drill holes.
 - If possible, use all the holes in the bracket, but at the very least use the lowest hole and the two that are furthest apart.
 - Screw the bracket firmly into place using suitable screws.
2. Hook the cooling channels of the inverter above onto the tabs on the bracket.
3. Set the SINVERT PVM inverter onto the edge of the bracket. Lift the inverter so you can reach the insert nut and push the inverter all the way to the wall.
4. Screw the SINVERT PVM inverter with the screws from the accessory pack (M5x20) from underneath onto the insert nuts in the bracket.
If a screw is lost, use only stainless steel screws to replace it.

CAUTION

Safety hazards associated with using unsuitable PV modules

The SINVERT PVM inverter is not suitable for use with PV modules for which pole grounding is an essential requirement. Modules of this type must not be connected to the SINVERT PVM inverter. Failure to provide suitable grounding poses a safety risk to personnel and the installation itself.

NOTICE

Loss of degree of protection

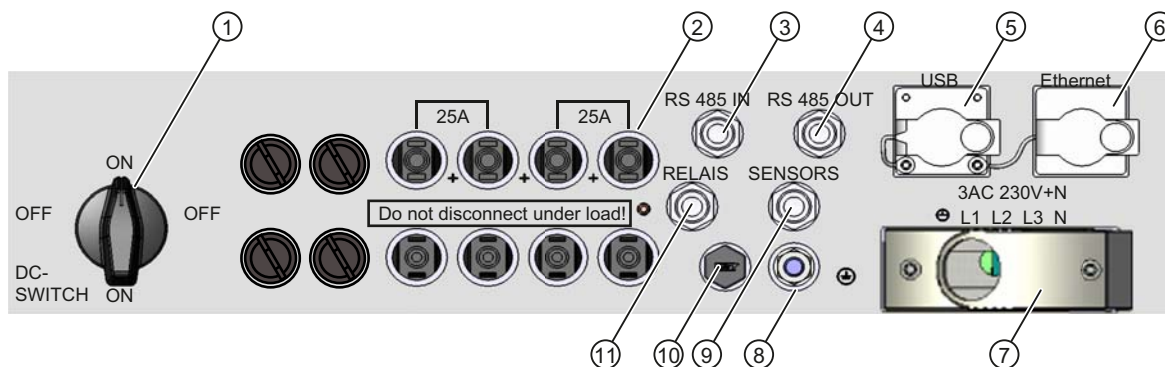
The IP65 degree of protection is retained only if you connect the connector from the accessory pack correctly. If you use different connectors, or the connectors are not properly connected, you lose the IP65 protection. Close all unused openings with blind plugs.

6.1 Device connection

The figure below shows the connections of the SINVERT PVM inverter on the underside of the device.

Device connection for SINVERT PVM10, SINVERT PVM13, and SINVERT PVM17 4DC ¹⁾

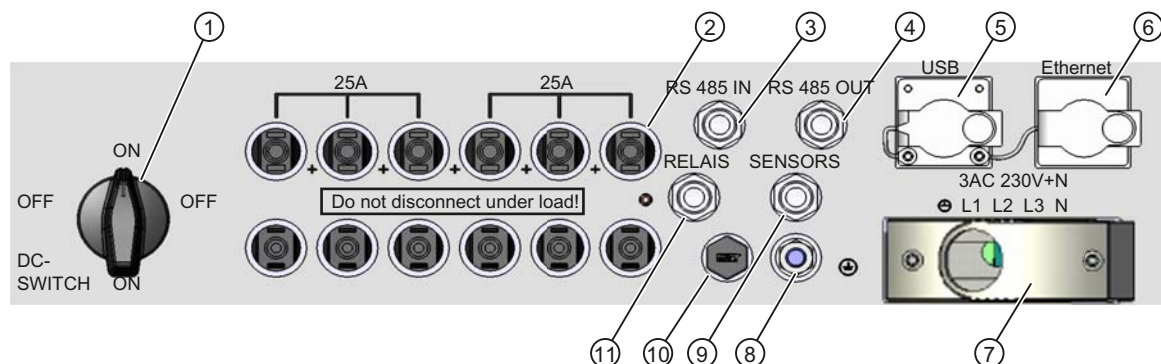
¹⁾ SINVERT PVM17 4DC inverter is no longer available to order.



- ① DC circuit breaker
- ② 4 pairs of PV generator connections: MC4 connector
Top row: +, bottom row: -
- ③ RS485 connection: IN
- ④ RS485 connection OUT
- ⑤ USB interface connection
- ⑥ Ethernet interface connection
- ⑦ Power connection
- ⑧ Ground connection M8
- ⑨ Connection for insulation sensor
- ⑩ Overpressure compensation
- ⑪ Connection for RELAY 230 V AC / 4 A (fault signaling)

Figure 6-1 Device connection for SINVERT PVM10, SINVERT PVM13, and SINVERT PVM17 4DC

Device connection for SINVERT PVM17 6DC and SINVERT PVM20



- ① DC circuit breaker
- ② 6 pairs of PV generator connections: MC4 connector
Top row: +, bottom row: -
- ③ RS485 connection: IN
- ④ RS485 connection OUT
- ⑤ USB interface connection
- ⑥ Ethernet interface connection
- ⑦ Power connection
- ⑧ Ground connection M8
- ⑨ Connection for insulation sensor
- ⑩ Overpressure compensation
- ⑪ Connection for RELAY 230 V AC / 4 A (fault signaling)

Figure 6-2 Device connection for SINVERT PVM17 6DC and SINVERT PVM20

6.2 Power connection



WARNING

Electric shock and fire hazard due to high leakage current

Create a ground connection before connecting to the PV field and the power supply circuit.

The power supply cable must be equipped with a suitable line protection: 32 A rated current, characteristic B. The maximum rated current is 29 A per phase. Reduction factors in wiring miniature circuit breakers in series must be taken into account. In doing so, the following standards must be observed:

- IEC 60364-5-523
Current-carrying capacities in wiring systems
As well as the relevant local standards such as DIN VDE 0298-4 for Germany
- IEC 364-4-43
Erection of power installations with rated voltages below 1000 V
IEC 364-4-473
Protective measures; protection of cables against overcurrent
As well as the relevant local standards such as DIN VDE 0100; Part 430 for Germany
- IEC 60364-4-41
Protective measures and protection against electric shock
As well as the relevant local standards such as DIN VDE 0100; Part 410 for Germany

In addition, the following points from the local power supplier should be observed:

- The relevant technical and specific rules
- There must be approval for the installation



WARNING

Danger of fatal electric shock

Before connecting the SINVERT PVM inverter to the AC power supply, the power connection must be switched off, the absence of power must be determined, and a safeguard must be provided to prevent it being switched on again.

CAUTION

An appropriate overload protection must be available.

1. Check the supply voltage. The voltage must not be higher than 265 V (phase to neutral conductor). If the line voltage is higher, the local power company must provide assistance.

2. Connect the SINVERT PVM inverter to ground using the indicated grounding bolt.
3. Insert the power cord connector on the supplied plug, following the diagram. Connect the power connector on the SINVERT PVM inverter and screw the plug firmly into position.

When using end sleeves with insulation, ensure the insulation of the end sleeves is not crushed.

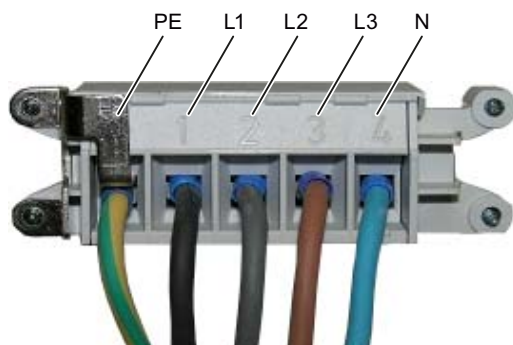


Figure 6-3 Power connection

6.3 Choice of power supply cable

Select the cross-section of the power supply cable in such a way that the line losses are as low as possible.

- The following points must be kept in mind:
 - A finely stranded cable is recommended as the feeder conductor for all cross-sections as it is most suitable.
 - The cable gland of the supplied connector housing permits a maximum external diameter of the supply cable of 18 mm. This results in supply cables of up to 5 x 6 mm² (e.g. Lapptherm 145, 5 x 6 mm²).
 - You can order a larger connector housing optionally that allows a maximum external diameter of the cable of 24 mm. This allows you to connect a cable of 5 x 10 mm². See section Spare parts and accessories (Page 89)
 - The terminals allow a maximum connecting cable cross-section of 10 mm².

Connector	Terminal area
AC connector from the accessory pack	External diameter: 10 ... 18 mm
AC connector available as an accessory	External diameter: 14 ... 24 mm

NOTICE

Damage to the connecting cable

Please note that the supplied connector housing only permits cable diameters up to 18 mm.

For a voltage drop $\leq 1\%$ in the cables, the following guide values for cross-sections and cable lengths typically apply.

Cable cross-section	4.0 mm ²	6.0 mm ²	10.0 mm ²
Maximum cable length	20 m	30 m	50 m

Note

To guarantee the degree of protection IP65, the connecting plugs and cables must be matched to each other and all unused connections must be provided with blanking plugs.

1. Tighten the cable gland to prevent the ingress of moisture.

6.4 Grounding

WARNING

Risk of electric shock

The PVM SINVERT inverter must be grounded on a threaded bolt. Otherwise a potential difference can build up, causing an electric shock hazard.

NOTICE

Possible EMC problems caused by inadequate grounding

In order to satisfy EMC standards, ground the inverter at the M8 grounding point with at least 10 mm²!

There is a threaded M8 grounding bolt on the connector side of the SINVERT PVM inverter. Grounding can also be considered an overvoltage protection measure. Therefore, for grounding, choose a cable cross section that is larger than the cross section of the supply line (at least 10 mm²). In addition, the ground cable should be installed at the greatest possible distance to the mains supply line and not directly parallel to it.

Correctly installed grounding of the module frame can be used as equipotential bonding for the abovementioned overvoltage protection.

Even if the AC connector is disconnected, the unit may still have live DC voltage. The unit would then not be grounded without the connection to the ground via the grounding bolt.

The unit must be connected to a TN-S system where possible. When it is connected to a TT system without separately run protective conductor PE, a separate PE cable with cross-section of at least 10 mm² must be connected to and grounded on the inverter.

6.5 Residual current protection

Note

The SINVERT PVMs are transformerless photovoltaic grid-connected inverters that meet the requirements for fault protection in accordance with IEC 60364-7-712. They also comply with the local standards DIN VDE 0100-712 for Germany and CEI 64-8/7 for Italy.

SINVERT PVM inverters ensure that there is no DC offset. SINVERT PVM inverters can therefore be operated with a Type A residual current device (RCD, FI) without affecting the function of either the protection or the inverter.

The rated fault current should be at least 100 mA per inverter.



WARNING

Risk of electric shock

Each inverter must be separately protected by an RCCB. To protect a group of inverters by a single RCCB with a higher fault current rating could have potentially fatal consequences and is therefore prohibited.

6.6 DC connection of the PV strings

Before connecting the PV strings, the device must be connected to the line in order to connect it **safely to PE**.

WARNING

Active PV strings can carry hazardous voltages!

The open-circuit voltage must be checked before connecting the PV strings. The open-circuit voltage must not exceed 50 V.

The device itself is not damaged if DC voltage is applied when the PV strings are connected.

- Check the PV strings to be connected for a possible ground fault. Rectify any existing ground fault before connecting the PV strings to the inverter.
- Ensure correct polarity when connecting the PV strings. If individual strings are connected incorrectly, module lines can be damaged. The SINVERT PVM inverter is protected by a reverse polarity protection diode. Secure the connections against unintentional disconnection.
- The following types of inverter have four inputs for the PV field:
 - SINVERT PVM10
 - SINVERT PVM13
 - SINVERT PVM17 4DC (no longer available to order)

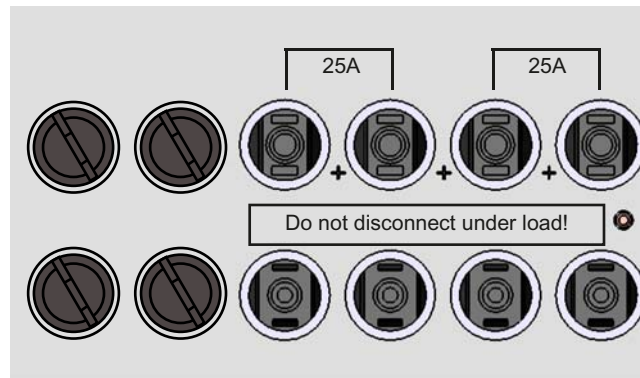


Figure 6-4 PV field connection with four DC inputs

- The following types of inverter have six inputs for the PV field:
 - SINVERT PVM17 6DC
 - SINVERT PVM20

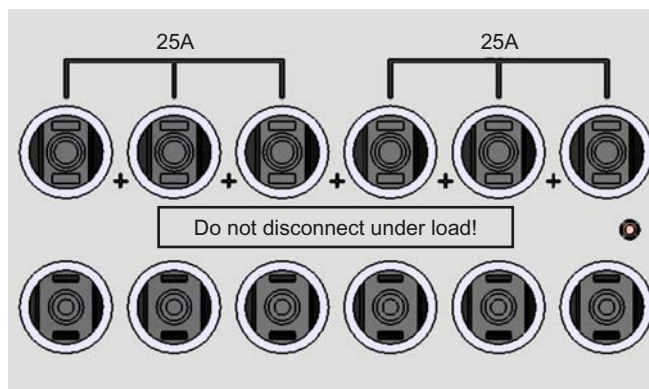


Figure 6-5 PV field connection with six DC inputs

- Close unused connections with blanking plugs. Otherwise the lower limit of the degree of protection (IP65) may be violated!
- Use DC connecting plugs that are suitable for the cable diameter. If cable connectors are used that are not suitable for the cable diameter, degree of protection IP65 is not guaranteed!

6.7 Selection of the DC connecting cable

The PV generator is connected via the plugs and sockets built into the housing. Cable cross-sections up to 6 mm² are possible.

MC4 connectors usually allow a cable cross-section of 6 mm².

CAUTION

Degree of protection IP65 only when using MC4 connectors

Degree of protection IP65 is only retained if MC4 connectors and MC4 sockets are used and correctly connected.

1. Connect the plugs and sockets in accordance with the connector manufacturer's installation instructions.

6.8 Insolation sensor

An insolation sensor is available as an optional accessory. You need a corresponding connector to connect the insolation sensor. The ordering data for the insolation sensor and the corresponding connector are listed in Section Spare parts and accessories (Page 89).

Table 6- 1 Terminal assignment

Color	Signal
Orange	Measuring signal insolation (0 to 10 V)
Brown	Measuring signal temperature (0 to 10 V)
Red	Supply voltage (12 to 24 V DC)
Black	GND

NOTICE

Damage to the device or malfunction possible

The shield of the sensor cable must be applied to PIN 2 and PIN 5, otherwise malfunctions or damage to the device may occur.

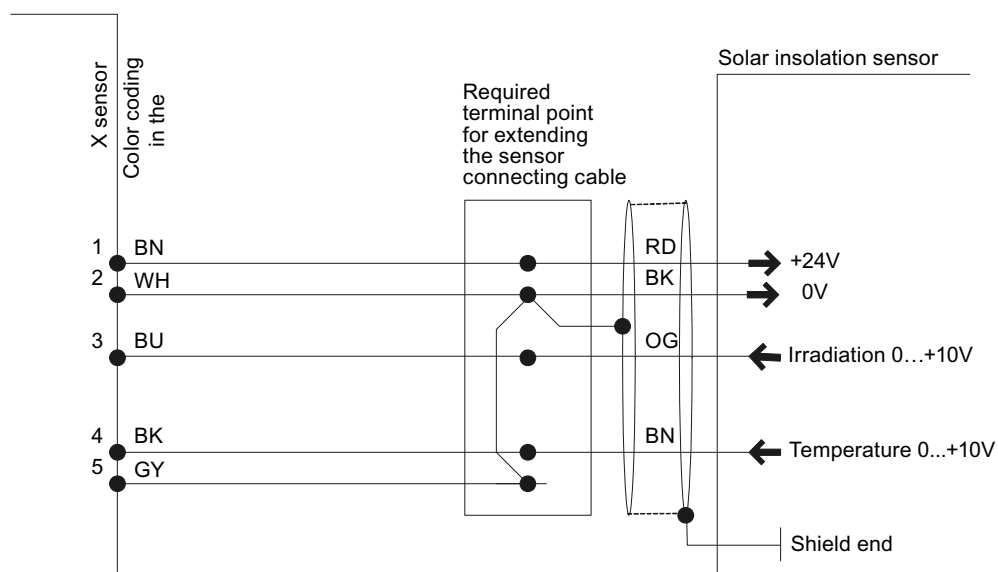


Figure 6-6 Connection of the insolation sensor

6.9 Interface connection

RS485

The RS485 connectors are included in the accessory pack.

RS485 out		RS485 in	
Pin 1	Bus terminator +	Pin 1	Reference +
Pin 2	RS485+ out	Pin 2	RS485+ in
Pin 3	RS485- out	Pin 3	RS485- in
Pin 4	Bus terminator -	Pin 4	Reference -

The RS485 interface supports the USS (Universal Serial Interface) protocol that can be used for data transfer, for example, to a datalogger or a remote monitor.

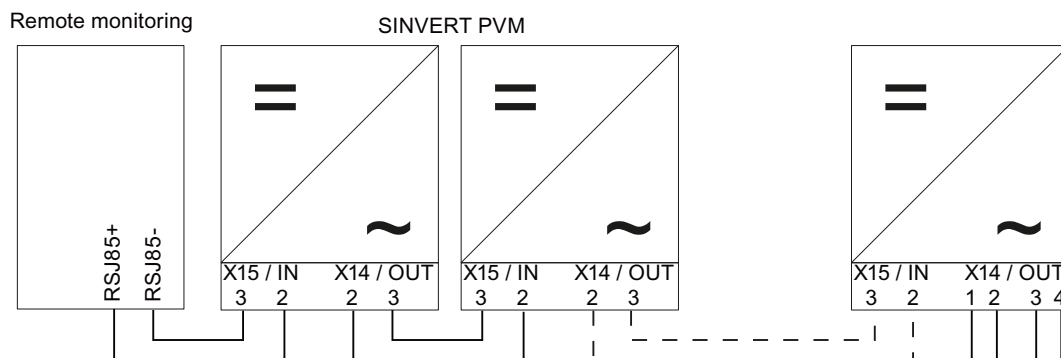


Figure 6-7 Connection of the standard interface

When operating this interface, you must ensure that every bus node receives a unique address, and that the correct data transfer rate is set with the relevant USS protocol.

Bus terminator

The bus must be terminated with 120 ohms at the RS485 out connection of the last bus node. You can use either the internal 120-ohm resistor or an external 120-ohm resistor for this purpose.

- Bus terminator with internal 120-ohm resistor:
Short-circuit the following pins with two wire bridges:
 - RS485 out Pin 1 with RS485 out Pin 2
 - RS485 out Pin 3 with RS485 out Pin 4
- Bus terminator with external 120-ohm resistor:
An external 120-ohm resistor is not included in the scope of supply.
Connect a 120-ohm resistor between RS485 out Pin 2 and RS485 out Pin 3.

USB

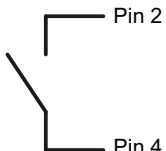
Examples of prefabricated IP20 cables on connector IP67 (not included in the scope of supply):

- Phoenix Contact VS-04-2X2X26C7/7-67B/SDA, Part No. 1653922 for 5 m
- Phoenix Contact VS-04-2X2X26C7/7-67B/SDA, Part No. 1653919 for 2 m

Ethernet

One connector for degree of protection IP67 is included in the accessory pack.

6.10 Relay connection

X61		
	Pin 1	Not assigned
	Pin 2	Contact capacity 4 A / 250 V AC-3
	Pin 3	Not assigned
	Pin 4	Contact capacity 4 A / 250 V AC-3

- The relevant connector is included in the accessory pack.
- The relay is programmed as a fault signal:
 - The contact closes when a fault occurs.
 - The contact opens when the cause of the fault has been removed or there is no DC voltage.

Commissioning

7.1 Prerequisites

The following activities must be completed before commissioning the SINVERT PVM inverter:

- Correctly installed power connection
- Correct installation of PV lines
- The connections are protected against unintended disconnection

WARNING

Risk of electric shock

- Before switching on, ensure the connectors are seated properly.
- Remove the PV generator connectors only when the following conditions have been met:
 - Switch the DC disconnect on the SINVERT PVM to "OFF".
 - Measure at the DC end to make sure that no current is flowing.
 - Disconnect the power supply and take measures to prevent the power being switched back on again.

WARNING

Electric shock and fire hazard due to high leakage current

Create a ground connection before connecting to the PV field and the power supply circuit.

7.2 Switching on the unit

1. Make sure that power is applied to the unit. This is accomplished by inserting the external line fuse or switching on the circuit breaker.
2. Put the DC disconnecter on the SINVERT PVM inverter in the ON position.
The inverter will only start up when the DC connector is switched on with the PV field connected.

NOTICE**Operator panel is only active with DC power switched on**

The operator panel with status indicators, display, and control keys is active only when DC voltage is switched on, since the electronics of the SINVERT PVM inverter are supplied exclusively on the DC side.

- The SINVERT PVM inverter starts at a DC voltage of approximately 200 V.
 - Status indicators, display and control keys are active.
 - The SINVERT PVM inverter is conditionally ready to run.
- At a DC voltage of about 350 V or over, the inverter starts to feed energy into the AC grid.

7.3 Selecting the country setting and menu language

The country setting determines the country-specific line monitoring parameters. In selecting the country setting, the menu language is set automatically. The menu language is then independent of the country setting and can be freely selected at any time using the menu.

No country setting is selected when the device is delivered.

CAUTION**The selected country setting can be changed by service personnel only!**

After selecting and accepting the country setting, you can no longer change the country setting yourself. According to the regulations, the country setting can be changed by service personnel only.

If the unit has been in operation, the country setting is already selected. In this event, you will not be asked to enter the country setting. The country setting can be changed by service personnel only.

CAUTION**Withdrawal of operating permit**

If you operate SINVERT PVM with an incorrect country setting, the electrical utility can withdraw your operating permit.

Commissioning is prohibited until the total system conforms to the national regulations and safety rules of the application.

We assume no responsibility for an incorrect country setting.

Selecting the country setting

Immediately after switching on the DC power, the following window appears on the display and prompts you to select the country setting. You can choose among the countries listed. The term "country setting" itself does not appear in the menu.

Note

Special features of the country settings for Germany and Italy

There are two country settings for Germany:

- "Deutschland ENS"
To feed into the low-voltage grid
- "Deutschland MSR"
To feed into the medium-voltage grid via a medium-voltage transformer

The "Italia Option" setting can be selected with special permission from ENEL, if difficult line conditions prevail at a site in Italy.

There are two country settings for Germany:

- Deutschland ENS
To feed into the low-voltage grid
- Deutschland MSR
To feed into the medium-voltage grid via a medium-voltage transformer

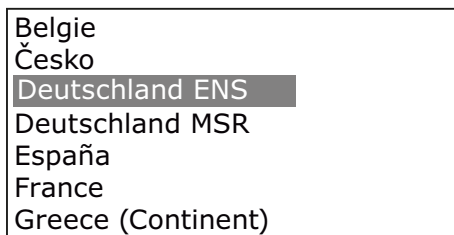


Figure 7-1 Selecting the country setting

1. Choose the country-specific country setting for your location using the "↓" and "↑" keys.
 - By selecting the country setting, you also select the menu language.
 - The menu language can be changed at any time in the menu.
2. Confirm selection with the "ENTER" key

Accepting the country setting

CAUTION
The selected country setting can be changed by service personnel only! For safety, a prompt appears asking whether you want to accept the choice of country setting. After accepting the country setting, you can no longer change the country setting yourself.

Accept ?
Yes = Enter
No = Esc

Figure 7-2 Accepting the country setting

1. Only confirm the country setting if you are certain.
 - If you are not certain, abort the process by pressing "ESC". In this case, you cannot operate the unit, and you cannot continue to use the menu.
 - If you are certain you wish to accept the country setting, confirm it with "ENTER".

Changing the menu language

The choice of language has no effect on the country setting. To change the menu language, proceed as follows:

1. Press the "F1" key to call the menu.
2. Choose the fourth menu item with the "↓" and "↑" keys: configuration.

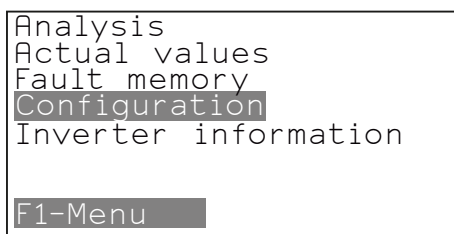


Figure 7-3 Menu item: "Select configuration"

3. Confirm selection with the "ENTER" key.
4. Choose the first menu item with the "↓" and "↑" keys: Languages.

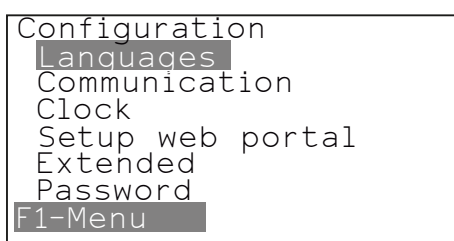


Figure 7-4 Menu item: "Choose language"

5. Confirm selection with the "ENTER" key.
6. Choose the desired menu language with the "↓" and "↑" keys.
7. Confirm selection with the "ENTER" key.

The menu switches to the desired language.

The display is initially empty.

8. Press the "ESC" button to return to the menu.

7.4 Activating the unit

1. Make sure that power is applied to the unit. This is accomplished by inserting the external line fuse or switching on the circuit breaker.
2. Put the DC disconnecter on the SINVERT PVM in the ON position.

Initialization and power-up procedure

Assuming that the PV field is receiving sufficient sunlight and there is no error present, the process described below should occur, which you can follow on the display.

The table provides an overview of the status display process on successful initialization and powering up.

Status	READY	ON	ALARM	GROUND FLT
Off	off	off	off	off
Self-test (approx. 6 s)	is lit	is lit	is lit	is lit
Initialization	flashes	off	off	off
Initialization complete	is lit	off	off	off
Switching in the inverter	is lit	flashes	off	off
Infeed operation	off	is lit	off	off

This is described, showing the screen display, in the sections below.

Activation code

The device has an activation code.

- The activation code is used to assign the SINVERT PVM inverter to the SINVERT Webmonitor web portal.
- The activation code is not required for commissioning the inverter.
- You will find more information about communication with the web portal in the operating instructions for the Webmonitor and in Chapter Communication (Page 59) (specifically in Section Assignment in the web portal (activation code for the SINVERT Webmonitor) (Page 63)).

Self-test

- Status display
All LEDs light up for approximately 6 seconds.

Initialization

The initialization operation is started. The system is checked during the prescribed waiting time:

- Status display
 - The **READY** LED flashes
- Display:
 - P_{AC} : Infeed power in watts (W)
 - U_{AC} : Line voltage in volts (V)
 - U_{DC} : PV field voltage in volts (V)
 - Status: Initialization

The initialization operation is completed.

- Status display
 - The **READY** LED is lit
- Display:
 - P_{AC} : Infeed power in watts (W)
 - U_{AC} : Line voltage in volts (V)
 - U_{DC} : PV field voltage in volts (V)

Switching in the inverter

The inverter starts up with a PV field voltage > 350 V.

- Status display
 - The **READY** LED is lit
 - The **ON** LED flashes
- Display:
 - ON
 - U_{DC} : PV field voltage in volts (V)
 - Activation

This procedure can take several minutes at initial startup. During normal operation, the procedure takes no more than 3 minutes.

Infeed operation

- Status display
 - The **ON** LED is lit
 - The **READY** LED is off
- Display
 - P_{AC}: Infeed power in watts (W)
 - U_{AC}: Line voltage in volts (V)
 - U_{DC}: PV field voltage in volts (V)
 - E Day: Daily energy
 - Operation

7.5 Checking the time of day

If the electronic circuitry has been without supply voltage for an extended period (approximately 2 to 3 weeks), the correct time of day may no longer be set.

Check the time of day and set the time as follows if required:

1. Call up the menu with "F1"
2. Select the menu item "Set clock" with the "↓" key
3. Use the "→" and "←" keys to select Day, Month, Year, Hour, Minute and Second in sequence, and set the value with "↑" and "↓"
4. Confirm your input with "ENTER"

7.6 Communication

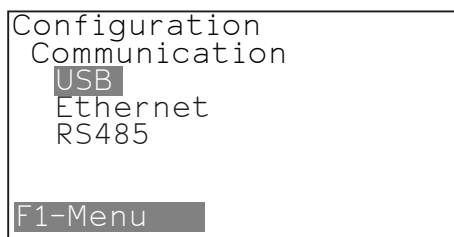


Figure 7-5 Communication

The Communication menu is used for configuring the USB, Ethernet and RS485 interfaces.

You will find an introduction to using the SINVERT PVM menu in Chapter Operator input (Page 67).

NOTICE

Parameter settings are only applied after the inverter has been switched off and then back on again.

7.6.1 Password

A password is required to change parameters. The customer password is 72555.

1. Enter the customer password under "Configuration/Password": "72555".

– Enter the digits from right to left:

- 5
- 5
- 5
- 2
- 7

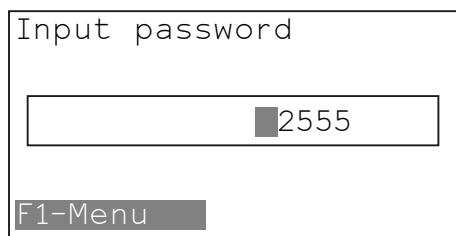


Figure 7-6 Display: Enter password

Parameter	Description
P2000	Password

7.6.2 Connecting a SINVERT PVM to the Internet portal

7.6.2.1 Setting the IP address and standard gateway

NOTICE

Parameter settings are only applied after the inverter has been switched off and then back on again.

Entering the IP address of the inverter

1. Enter the required IP address of the inverter under "Configuration/Communication/Ethernet/IP address".
 - The factory setting is "192.168.0.123".

Example: IP address 192.168.0.123

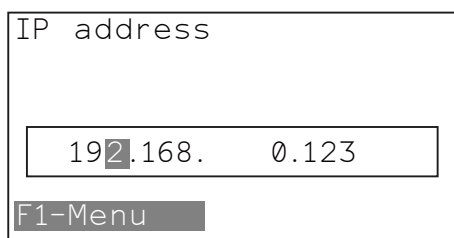


Figure 7-7 Display: Enter IP address

Parameter	Description	Value (example)
P0410.00	IP address: 1st number	192
P0410.01	IP address: 2nd number	168
P0410.02	IP address: 3rd number	0
P0410.03	IP address: 4th number	123

Setting the subnet mask

You do not usually have to change the subnet mask. The factory setting is 255.255.255.0. You can change the setting under "Configuration/Communication/Ethernet/IP subnet mask".

Example: Subnet mask: 255.255.255.0

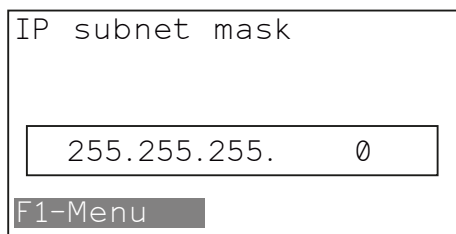


Figure 7-8 Ethernet: Enter subnet mask

Parameter	Description	Value (example)
P0411.00	Subnet mask: 1st number	255
P0411.01	Subnet mask: 2nd number	255
P0411.02	Subnet mask: 3rd number	255
P0411.03	Subnet mask: 4th number	0

Setting the standard gateway

1. Enter the IP address of the router in your LAN under "Configuration/Communication/Ethernet/Standard gateway".
 - The factory setting is "192.168.0.1".

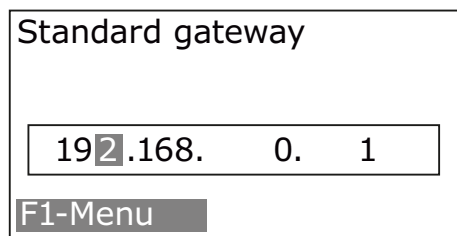


Figure 7-9 Standard gateway

Parameter	Description	Value (example)
P0414.00	Standard gateway: 1st number	192
P0414.01	Standard gateway: 2nd number	168
P0414.02	Standard gateway: 3rd number	0
P0414.03	Standard gateway: 4th number	1

Activating the "Data transfer to portal" option on the inverter.

You can use the "Portal settings" menu command to set whether data is to be sent to an external portal, e.g. the SINVERT Webmonitor. The portal settings are deactivated on delivery.

1. Set the value 1 under "Configuration/Portal settings/Activation".
 - The value is factory-set to 0.

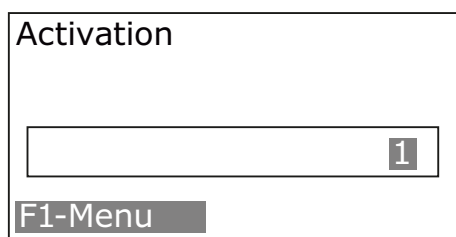


Figure 7-10 Portal settings

Parameter	Description	Value (example)
P0473	Portal settings 0: Portal settings off 1: Portal settings on	1

IP address assignment when using more than one inverter

1. If you are using more than one SINVERT PVM inverter, repeat the above steps for each of them.
 - Each inverter in the system must be assigned a dedicated IP address which has not previously been assigned.

Internet access to the inverter

The procedure for setting up Internet access to the portal and adding one or more inverters is described in the operating instructions for the SINVERT Webmonitor.

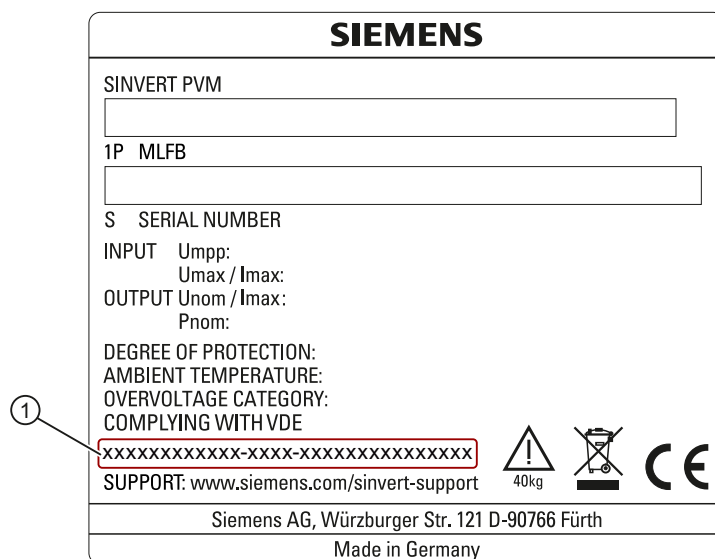
You will find the portal itself at:

SINVERT Webmonitor (<https://www.automation.siemens.com/sinvertwebmonitor>)

7.6.2.2 Assignment in the web portal (activation code for the SINVERT Webmonitor)

You need an activation code to assign the SINVERT PM in the SINVERT Webmonitor web portal. The SINVERT Webmonitor documentation contains more detailed information about this.

The activation code valid for your inverter is indicated on the SINVERT PVM rating plate (see the figure below). The rating plate is located on the right-hand side of the inverter. The activation code is not required for commissioning the inverter.



① Position of the activation code for the SINVERT Webmonitor

Figure 7-11 Rating plate with activation code

7.6.2.3 Using other web portals

We recommend that you use the Siemens SINVERT Webmonitor web portal. You can also use other web portals such as SolarLog or MeteoControl for SINVERT PVM inverters. The factory setting is the SINVERT WebMonitor.

NOTICE

If you are using a SINVERT PVM Control Box you can only use the SINVERT Webmonitor web portal.

Monitoring using SolarLog or MeteoControl

To use SolarLog or MeteoControl to monitor SINVERT PVM inverters, proceed as follows:

This procedure also applies if you wish to switch from SolarLog or MeteoControl to the SINVERT Webmonitor.

1. Press the "F1" key to call the menu.
2. Select the menu command "Configuration" with the "↓" and "↑" keys and press "ENTER".
3. Select the "Password" menu command and press "ENTER".
4. Enter the customer password 72555.
 - Enter the numbers from right to left: 5 5 5 2 7 and confirm with "ENTER".
5. Select the "Communication" menu command and press "ENTER".
6. Select the "RS485" menu command and press "ENTER".
7. Select the "USS address" menu command and press "ENTER".
8. Enter the required address for the inverter (1 to 31) and confirm with "ENTER".
 - We recommend setting consecutive addresses starting at 1, i.e. 1, 2, 3, etc.
 - The highest possible address is 31.
9. Select the "Protocol" menu command and press "ENTER".
10. Select the communication method for the required web portal. To do this, enter the numbers representing the communication method and confirm with "ENTER".
The communication methods for the following web portals must be different:
 - SINVERT Webmonitor: 1
 - SolarLog: 2
 - MeteoControl: 3
11. Press "ESC" four times to access the power rating display.
12. To apply the settings, switch the SINVERT PVM off and then back on again.
13. Switch the SINVERT PVM off at the DC disconnect, wait about a minute, and then switch it back on again.

7.6.3 RS485

Used for configuring the RS485 interface.

Parameter	Description
P0405.00-05	Service level
P0406.00-05	RS485 address 0-31. The address is freely selectable. Each device requires its own address – No double assignment! Otherwise, assignment for the SINVERT Webmonitor is not possible and functions controlled via RS485 are no longer available.
P0407.00-05	RS485 protocol

7.6.4 USB

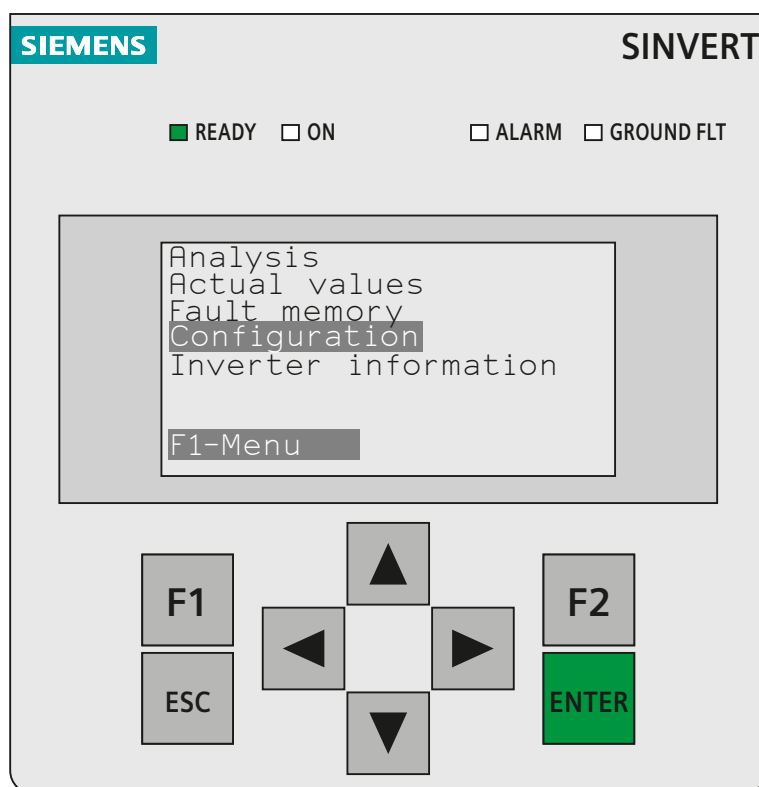
USB

Used for configuring the USB interface.

Operator input

8.1 Status display and operator controls

Operator controls



Meaning of the operator controls

F1	Calling up the device menu
F2	Select entry of normalized energy data
↑ ↓	Selection
← →	Function in the menu: Jump to first or last menu item
	Function on parameter change: Position left, position right (skip screen)
ESC	Fault acknowledgement, entry deletion
ENTER	Confirmation of menu selection and input

Figure 8-1 Operator panel

You will find further information on the operator panel in the section Operator panel (Page 21)

Status display

The start screen is the status display. From here, you can call up the device menu and change to displaying the infeed power.

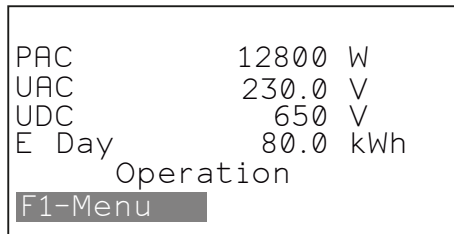


Figure 8-2 Display: Operation

P_{AC}: Instantaneous infeed power in watts (W)

U_{AC}: Line voltage in volts (V)

U_{DC}: PV field voltage in volts (V)

E Day: Daily energy in kWh

8.2 Overview of menu navigation

The following overview shows how you can go from the status display direct to the display of the energy data and infeed power level with the keys "←", "→", and "F2".

The "F1" key takes you to the device menu. In the device menu, choose menu items with the "↑" and "↓" keys, open submenus with "ENTER", and confirm entries with "ENTER". The "ESC" key cancels entries and takes you back to the next higher menu level.

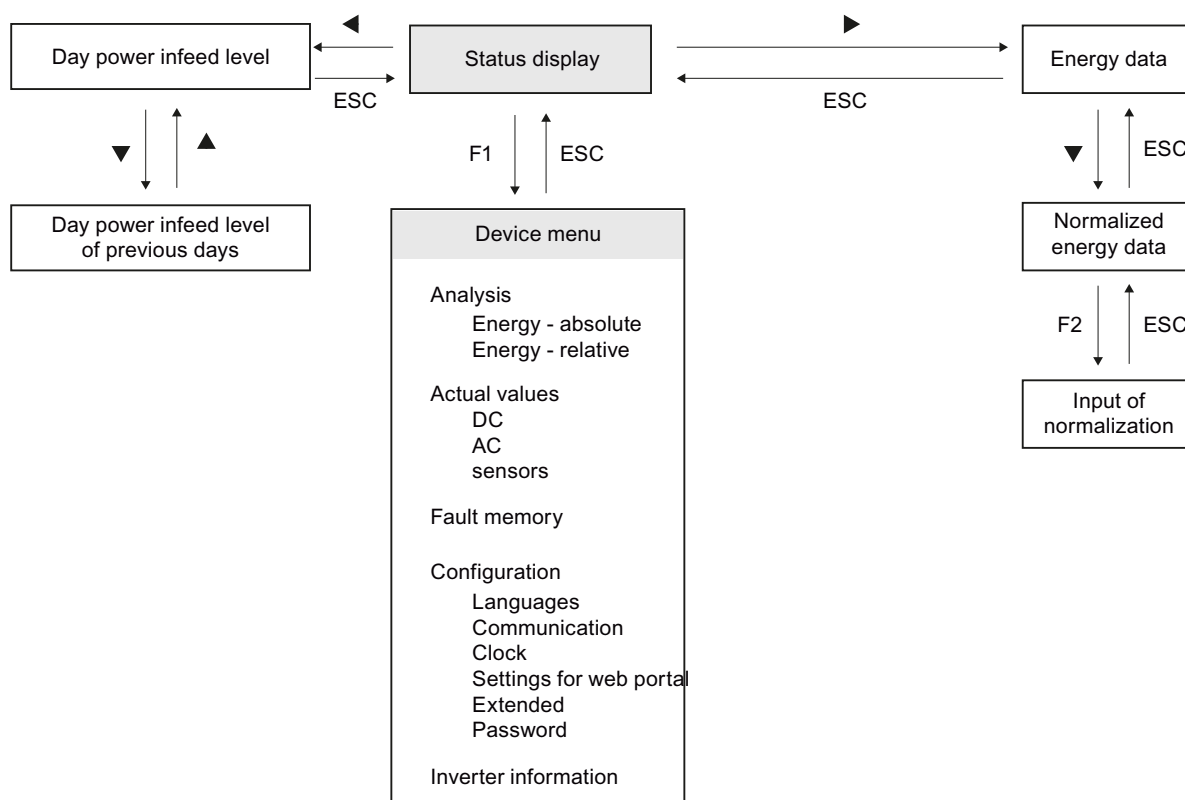


Figure 8-3 Menu navigation

8.3 Display of the infeed power and the energy data

Graphical display of the infeed power

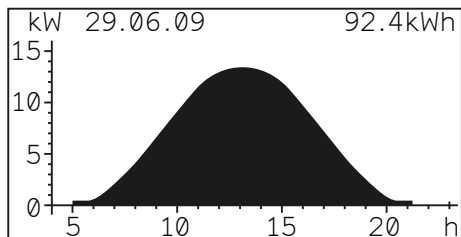


Figure 8-4 Display: diagram

Day infeed power level

To display the trend of the day infeed power level, press "←" once.

Infeed power level of previous days

To display the trend of the infeed power level for previous days, press "↓".

Return to the status display

To return to the status display, press "ESC".

Displaying the energy data

To show the current energy data, as well as the operating hours accumulated to-date, press "→" once.

Displaying the normalized energy data

To display the trend of the normalized energy data, press "→" twice.

You can enter the actual connected generator power ("Power normalized") under the menu item "Numerical list" at Parameter P1155.

Return to the status display

To return to the status display, press "ESC".

Input of normalization

To see the normalized energy data, press "F2".

Enter the connected PV generator power as follows:

Key "←": Selecting the integer position

Key "→": Selecting the decimal position

Key "↑": Digit of the selected position is incremented by 1 each time the key is pressed.

Key "↓": Digit of the selected position is decremented by 1 each time the key is pressed.

Return to the status display

To return to the status display, press "ESC".

Note**Direct input of parameter 1155**

If the input of normalization following the above procedure fails, you can also enter the value directly as parameter 1155 in the Configuration/Extended menu. For details, see Section Extended (Page 75).

8.4 Device menu

8.4.1 Overview

Overview of the device menu

To display the device menu, press "F1".

The following menu items are available:

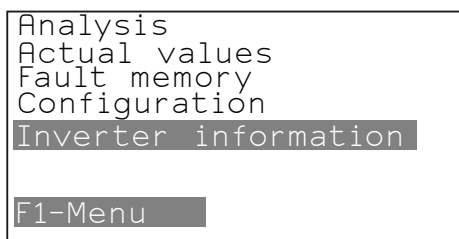


Figure 8-5 Display: Menu

Display of the parameters

Note

In various menu items, the display shows parameters starting with a "P" or "D".

- You can monitor and set values that start with a "P".
"P" stands for "programmable". Programmable
 - You can only monitor values that start with a "D".
"D" stands for "display". Can only be displayed
-

See also

Faults (Page 82)

8.4.2 Analysis

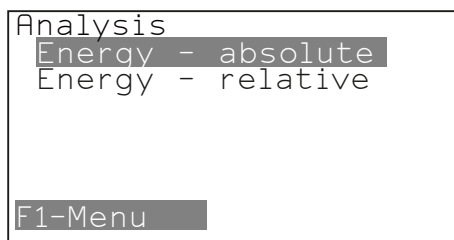


Figure 8-6 Analysis

You can choose whether the absolute energy or the relative energy is to be shown in the status display.

8.4.3 Actual values

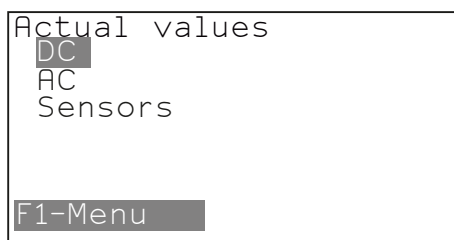


Figure 8-7 Actual values

Current measured values of the device. The status display also shows some of these values.

Parameter	Description
D1100	DC link voltage
D1104	Voltage of the PV field
D1105	Output current of the PV field
D1106	Fed-in line power (Watts)
D1107	PV field power (Watts)

8.4.4 Fault memory

```
Fault memory

00 System reboot
   14.10.2009 14:51:42
01 Frequency FLL
02 Frequency FLL
F1-Menu
```

Figure 8-8 Fault memory

Indication of whether there is a fault in the controller or the power stack. The fault message appears on the display. You can find more detailed information about current fault messages and about the fault message memory in section Faults (Page 82).

8.4.5 Configuration

```
Configuration
Languages
Communication
Clock
Setup web portal
Extended
Password
F1-Menu
```

Figure 8-9 Configuration

NOTICE
Parameter settings are only applied after the inverter has been switched off and then back on again.

Languages

The "Languages" menu is used for selecting the menu language. The choice of language has no effect on the country setting. For more details, see Section Selecting the country setting and menu language (Page 52).

Communication

The "Communication" menu is used for configuring the USB, Ethernet, and RS485 interfaces. For more details, see Chapter Communication (Page 59).

Date/time

The menu is used for setting the date and time. See Section Checking the time of day (Page 58).

Portal settings

You can use this menu command to set whether data is to be sent to an external portal, e.g. the SINVERT Webmonitor. The portal settings are deactivated on delivery. For more details, see Section Setting the IP address and standard gateway (Page 60).

Advanced

This menu contains menu commands for service purposes and is not significant for normal operation. For more details, see Section Extended (Page 75).

Password

A password is required to change parameters. For more details, see Section Password (Page 59).

8.4.5.1 Extended

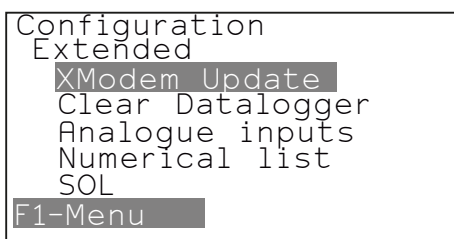


Figure 8-10 Extended

This menu contains menu items for service purposes and is not significant for normal operation.

Numerical list

Meaning of the parameters:

Parameter number	Parameter name	Unit
1107.0	DC power	Watt
1104.0	DC voltage	Volts
1105.0	DC current strength	Ampere
1106.0	AC power	Watt
1121.0	AC voltage peak L 1	Volts
1121.1	AC voltage peak L 2	Volts
1121.2	AC voltage peak L 3	Volts
1123.0	AC effective voltage average value	Volts
1141.0	AC current strength L 1	Ampere
1141.1	AC current strength L 2	Ampere
1141.2	AC current strength L 3	Ampere
1124.0	AC current strength: Total L1 + L2 + L3	Ampere
1122.0	AC frequency 1	Hertz
1122.1	AC frequency 2	Hertz
1122.2	AC frequency 3	Hertz
92.0	Heatsink temperature right	° C
92.1	Internal device temperature top left	° C
92.2	Internal device temperature bottom right	° C
92.3	Heatsink temperature left	° C
1191.0	Insolation sensor	Watt / m ²
1193.0	Temperature sensor ¹⁾	° C
501.0	Current status	-
500.0	Fault code	-
1150.0	Daily energy	kWh
1153.0	Monthly energy	kWh
1154.0	Annual energy	kWh
1151.0	Total energy	kWh
1155.0	Rated output of the PV field	kWp

¹⁾ Temperature sensor or temperature measurement of the insolation sensor, if available

8.4.6 Inverter information

Contains information about the device, e.g. the firmware version.

Service and maintenance

9.1 Maintenance

The SINVERT PVM inverters are maintenance-free.

- Check regularly that the cooling ducts on the rear of the inverter are free from coarse dirt.



Figure 9-1 Cooling ducts

- Do not use abrasive cleaning agents to clean the operator panel. Use a dry cloth or water.

9.2 Inverter replacement

Unpacking the replacement inverter

You have to return the old inverter in the packaging in which you receive the replacement inverter. Therefore, when unpacking the replacement inverter, take note of how the package is assembled.

Taking the inverter/the system out of operation

1. Set the DC switch on the underside of the inverter to "OFF".

Replacing the inverter



CAUTION

Risk of injury and damage to connectors caused by improper handling

If the connectors are not removed correctly they may sustain damage. The function of the inverter would no longer be assured.

- Do not pull on the cables.
- Use an unlocking tool for MC4 connectors to remove the connectors at the DC inputs. If you do not use the unlocking tool, you could injure yourself when removing the connectors and damage the connectors.

1. Remove all inverter connectors on the old inverter.
 - The connectors must be reconnected in the same way on the replacement inverter. Therefore, mark the connectors accordingly if necessary.
2. Unscrew the old inverter from the wall bracket.
 - To do this, loosen the two screws which attach the inverter to the wall bracket from below.
 - Please note that the SINVERT PVM inverter weighs approx. 40 kg.
3. Lift the inverter up and off the bracket.
4. Set the inverter down so that it cannot topple over or get scratched.
Damage to the inverter can incur additional costs.
5. Set the DC switch on the underside of the replacement inverter to "OFF".
6. Mount the replacement inverter on the wall bracket.
 - Follow the safety instructions and the information about how to proceed described in the steps below:
Protection during handling and installation (Page 34)
Installation of SINVERT PVM inverter (Page 35)

7. Connect the replacement inverter.
 - Make all of the connections as designated on the replacement inverter.
 - You will find more detailed information about inverter connection in the following chapter: Connecting (Page 37)

Commissioning the inverter

The settings on the replacement inverter must be aligned with your PV system.

1. Set the DC switch on the underside of the inverter to "ON".
2. Make the correct country setting.
3. Check the settings on the inverter.
 - You will find information about inverter commissioning in the following chapter: Commissioning (Page 51)

Activating the replacement device for the Webmonitor

1. Replace the activation code for the inverter in the Webmonitor.
 - For details, see Section Assignment in the web portal (activation code for the SINVERT Webmonitor) (Page 63).
- The data of the old inverter is replaced with the data of the replacement inverter.
- The historical statistical data of the old inverter is retained.
- If you do not update the activation code when you replace the inverter, data may be lost.

Returning the old inverter

1. Pack the old inverter carefully using the packaging in which the replacement device was received and in exactly the same way as it was received.
2. Send the old inverter back.
 - Your sales partner will provide the return address.

9.3 Disposal

Note

The SINVERT PVM inverter is RoHS-compliant. This means the device can be disposed of at local sites for the disposal of household devices.

Troubleshooting

10.1 Self-test - error messages

The system carries out a self-test following the initialization routine. The individual parts of the microcomputer system, such as the firmware and the data set, are checked here and the data read in by the power control board. Possible remedial measures can be derived from the type of fault.

10.2 Transient fault

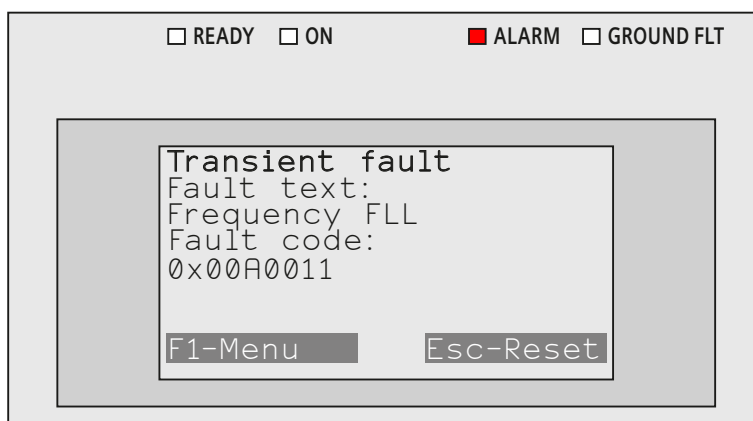


Figure 10-1 Transient fault

When certain faults occur, the inverter temporarily goes off-grid.

In contrast to faults, the "transient fault" is automatically acknowledged by the inverter and a new power-up attempt is made when the cause has been removed.

After a transient fault, the red status LED **ALARM** on the operator panel flashes. The associated error message is saved in the fault memory, safe from power outages. If there is a fault, the status LED **ALARM** is lit continuously. See section Faults (Page 82).

10.3 Faults

Permanently programmed and parameterizable limit values are continuously monitored during operation. To protect the power module of the SINVERT PVM inverter against damage, it is switched off-circuit when a limit value is exceeded or when a fault occurs. The appropriate error message is displayed.

In the event of a fault, the red status LED **ALARM** on the operator panel flashes.

Error messages are saved in the fault memory, safe from power outages. The fault memory is called via the display. The last 120 faults are stored in the fault memory. The last fault is stored in memory location S0, and the oldest is stored in S119. A new fault is always stored in memory location S0. All older faults are shifted up one location in the memory. The fault at memory location S120 is lost.

10.4 Isolation faults

In the event of an isolation fault, the **GROUND FLT** status LED lights up. The error message A010C INV appears on the display: ENS_ISO.



WARNING

Risk of electric shock and fire

An isolation fault must be remedied as soon as possible. The fault must only be remedied by electrical technicians.

The isolation resistance must be greater than 1.1 Mohms.

10.5 Acknowledging a fault

Following a shutdown on faults, restart of the device is locked until the fault has been acknowledged. While the cause of the fault is still active, acknowledgment is not possible. Only when the cause of the fault has been removed can the fault be acknowledged.

1. To acknowledge the error message, press "ESC" or switch the SINVERT PVM inverter off and on again at the DC switch.

10.6 Frequent error messages

Frequent error messages, their causes, and how to rectify them are listed below.

Error code	Error message	Cause	Elimination
000A0003	Controller voltage 3	Unbalance in lower DC link	Sporadic occurrences (1 to 2 times per day) indicate non-critical control disturbances caused by switching operations on the line, for example. If the problem occurs constantly, check the normalizations or insulation.
000A0004	Controller voltage 4	Unbalance in upper DC link	Sporadic occurrences (1 to 2 times per day) indicate non-critical control disturbances caused by switching operations on the line, for example. If the problem occurs constantly, check the normalizations or insulation.
000A0010	Undervoltage phase	Detection of line undervoltage on a phase conductor	The error is possibly caused by switching actions in the system. Remeasure line voltage, check fuses.
000A0011	ERROR_FLL	Voltage unbalance in the system Check the medium-voltage system if applicable.	Check the phases fuse/NH might have failed
000A0012	Frequency FLL	Inverter cannot detect any line voltage or frequency.	Check line voltage/frequency/fuses.
000A000E	Undervoltage	Inverter detects insufficient AC voltage.	Power failure; check AC voltage and correct if necessary.
000A000D	Overvoltage	Inverter detects excessive AC voltage.	Could be caused by switching actions on the line; check AC voltage and correct if necessary, check AC connectors, check country setting.
000A0100	Fault message LT Group fault alarm	Fault message from power unit	Further errors are queued. See fault memory.
000A0108	ENS_FREQ	Measured values outside tolerance	Check phases; fuse/NH might have failed
000A0111	Solar voltage PU	Solar voltage too high; too many PV modules are connected in series.	Check and if necessary correct system dimensioning.
000A010C	ISO fault	Inverter detects an ISO fault, ground fault on system.	Eliminate the ground fault, can also only occur due to moisture (connectors not sealed tight) or the system will only run after it has dried out.

Technical data

11.1 SINVERT PVM inverter

For inverter variants, see Section Introduction (Page 9).

DC data

Type	SINVERT				
	PVM10	PVM13	PVM17 4DC ¹⁾	PVM17 6DC	PVM20
Maximum DC power	10.2 kW	12.6 kW	16.8 kW	16.8 kW	19.6 kW
MPP tracking range (for P_{max})	380 to 850 V	420 to 850 V	525 to 850 V	460 to 850 V	480 to 850 V
Maximum DC voltage	1,000 V				
Maximum DC power	29 A	30 A	32 A	37 A	41 A
Maximum DC power per input	25 A				
Number of inputs for the PV generator	4	4	4	6	6
Internal overvoltage protection	Type 3 protection				

¹⁾ No longer available to order

AC data

Type	SINVERT				
	PVM10	PVM13	PVM17 4DC ¹⁾	PVM17 6DC	PVM20
Rated AC output	10.0 kW	12.4 kW	16.5 kW	16.5 kW	19.2 kW
Maximum AC power	10.0 kW	12.4 kW	16.5 kW	16.5 kW	19.2 kW
AC power connection	400 V 3 AC +N, 50 ... 60 Hz				
Cos phi	1; may be set up to 0.9 with capacitive or inductive component				
Maximum AC power	18 A	18 A	25 A	29 A	29 A
Fusing at AC end: Circuit breaker	32 A				
THDI	< 2,5%				
Maximum efficiency	98,0%			98,2%	
European efficiency	97,4%	97,5%	97,7%	97,8%	97,8%
Supply from	60 W				
Intrinsic consumption, night	< 0.5 W				
Internal overvoltage protection	Type 3 protection				

¹⁾ No longer available to order

Cooling, ambient conditions, EMC

Type	SINVERT				
	PVM10	PVM13	PVM17 4DC ¹⁾	PVM17 6DC	PVM20
Cooling	Natural convection				
Ambient temperature during storage and transport	- 25 to + 70 °C				
Ambient temperature during operation (with derating)	- 25 to + 55 °C				
Maximum ambient temperature for rated power	50 °C		45 °C		40 °C
Installation altitude	Up to 2,000 m above sea level				
Noise	< 45 dBa				
Emitted interference	EN 61000-6-4				
Manufacturer's declaration of conformity	CE				
Interference immunity	EN 61000-6-2				
Environmental classes	4K4H in accordance with IEC 60721-3-4				
ENS	In accordance with VDE0126-1-1				

¹⁾ No longer available to order

Mechanical properties

Type	SINVERT				
	PVM10	PVM13	PVM17 4DC ¹⁾	PVM17 6DC	PVM20
Degree of protection	IP65 in accordance with IEC 60529				
Dimensions Width/height/depth	530 mm / 601 mm / 270 mm				
Minimum distances Side/top/bottom	50 mm / 500 mm / 500 mm				
Weight	Approx. 40 kg			Approx. 41 kg	

¹⁾ No longer available to order

11.2 Insolation sensor

An insolation sensor is available as an optional accessory. You need a corresponding connector to connect the insolation sensor. The ordering data for the insolation sensor and the corresponding connector are listed in Section Spare parts and accessories (Page 89).

Type	Insolation temperature sensor Si-13TC-T-K	Insolation sensor Si-13TC-K
General		
Current measuring shunt	0.12 Ω (TK = 20 ppm/K)	
Ambient temperature	- 20 °C to + 70 °C	
Power supply	12 to 28 V DC	
Current consumption	0.3 mA	
Connecting cable	4 x 0.14 mm², 3 m (UV-stable)	
Cell size	50 x 33 mm	
External dimensions: Length/width/height	145 x 86 x 39 mm	
Weight	340 g	
Solar radiation (insolation)		
Measuring range	0 ... 1300 W/m²	
Output signal	0 ... 10 V	
Measuring accuracy	± 5% of the full-scale value	
Device temperature		
Measuring range	- 20 °C to + 90 °C	-
Output signal	2.268 V + T [°C] * 86.9 mV/°C	-
Measuring accuracy	+ 1.5% at 25 °C	-
Non-linearity	± 0.5 °C	-
Max. deviation	± 2 °C	-

Ordering data

12.1 SINVERT PVM inverter

Inverter

Designation	Order number (MLFB)
SINVERT PVM10	6AG3120-3JB02-0AC0
SINVERT PVM13	6AG3120-3JE02-0AC0
SINVERT PVM17 6DC	6AG3120-3JK02-0AC0
SINVERT PVM20	6AG3120-3JM02-0AC0

12.2 Spare parts and accessories

Spare parts

Spare part	Supplier	Product designation	Order number
Connector set (accessory pack)	Siemens AG		A5E02777937
4-pin RS485 connector	Phoenix Contact	SACC-M12MS-4SC SH	1513253
4-pin relay socket	Phoenix Contact	SACC-M12FS-4SC SH	1513279
Ethernet connector	Phoenix Contact	VS-08-RJ45-5-Q/IP67	1656990
Wall bracket	Siemens AG		A5E02777939

Accessories

Accessories	Supplier	Product designation	Order number
Large AC connector (10 mm ² connection)	Siemens AG		A5E02810965
Key for MC4 connector unlocking tool	MultiContact	PV-MS	32.6024
Insulation sensor	MenckeTegtmeyer	Si-13TC-K	Product designation is sufficient
Insulation temperature sensor	MenckeTegtmeyer	Si-13TC-T-K	Product designation is sufficient
Connector for insulation sensor	Phoenix Contact	SACC-M12MS-5SC SH	1512555
SINVERT PVM Control Box	Siemens AG		6AG3600-3AB10-0AA0

Appendix

A.1 Technical support

Technical support for SINVERT products

- Information material und downloads for SINVERT products:
SINVERT infocenter (<http://www.siemens.com/sinvert-infocenter>)
Here you can find, for example:
 - Catalogs
 - Brochures
- Documentation on SINVERT products:
SINVERT support (<http://www.siemens.com/sinvert-support>)
Here you can find, for example:
 - Manuals and operating instructions
 - The latest product information, FAQs, downloads, tips and tricks
 - Characteristics and certificates
- Contacts for SINVERT are available at:
SINVERT partners (<http://www.siemens.com/sinvert-partner>)

Technical assistance for SINVERT products

For all technical queries, please contact:

- Phone: +49 (911) 895-5900
Monday to Friday, 8 am – 5 pm CET
- Fax: +49 (911) 895-5907
- E-mail: Technical assistance (<mailto:technical-assistance@siemens.com>)

A.2 Latest documentation

The latest versions of the documentation for all SINVERT products can be found on the Internet by following this link:

SINVERT support (<http://www.siemens.com/sinvert-support>)

The available documents include:

- SINVERT PVM Operating Instructions (compact)
- SINVERT PVM Operating Instructions
- Inverter data sheets
- Data sheet for the SINVERT PVM Control Box
- SINVERT PVM Control Box Operating Instructions
- SINVERT Webmonitor Operating Instructions

Certificates

You can find the certificates for the SINVERT PVM inverters on the Internet:

SINVERT support (<http://www.siemens.com/sinvert-support>)

Glossary

ENS

Line monitoring device with allocated switching elements

EVU

Electrical utility

FLL

English: Frequency locked loop

Frequently locked loop

INV

Inverter

MPP tracking

English: Maximum power point tracking

MSR

Medium Voltage Directive

$P_{\max}$

Maximum output power

PV

Photovoltaic

RoHS

English: Restriction of the use of certain hazardous substances

EC Directive 2002/95/EC on the restriction of the use of certain hazardous substances in electrical and electronic devices

Sea level

Sea level

THD

English: Total harmonic distortion

Total harmonic distortion Non-linear distortion of an electrical or acoustic signal

THDI

THD for electrical current

TN-S system

French: Terre Neutre Separé

Low-voltage system with neutral conductor N connected separately and protective conductor PE connected separately.

TT system

French: Terre Terre

Low-voltage system without protective conductor PE connected separately

USS protocol

Universal serial interface protocol

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